

Niraj Bharambe

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SUMMARY

Software engineer with close to two years of professional experience building production-grade **Python** and **React + TypeScript** systems. Shipped full-stack products from zero to one — distributed platforms, AI pipelines, and real-time cloud infrastructure. Completing an M.Sc. in Computer Science at Trinity College Dublin (Sept 2026, predicted First-Class Honours), with a dissertation in **multi-agent deep reinforcement learning**. Comfortable owning hard problems end-to-end in fast-moving teams.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Go, Java, C/C++, SQL, Bash, Solidity

Frontend & Mobile: React 18, React Native (Expo), Next.js, Tailwind CSS, Redux, Zustand, Responsive Design, WCAG 2.1 AA

Backend & Data: FastAPI, Node.js, Express, REST APIs, gRPC, PostgreSQL, Redis, ClickHouse, RabbitMQ, SQLite

AI & ML: LangChain, RAG, Agent Frameworks, Vertex AI (Gemini Pro + Flash), PyTorch, TensorFlow/TFLite, XGBoost

Cloud & DevOps: GCP, GKE (Kubernetes), Docker, Apache Airflow, GCP Pub/Sub, Firebase, Git, GitHub Actions, CI/CD

EXPERIENCE

Software Developer

Oct. 2023 – Jul. 2025

Humanify Technologies

Mumbai, India

- Owned end-to-end delivery of a **React + TypeScript** SaaS research dashboard used by 50+ researchers daily; architected **Redux** state management, reusable component libraries, and **REST API** integration — cutting reporting turnaround by 30%.
- Architected a production **Python + LangChain** agentic RAG pipeline over 5,000+ sources; elevated AI output quality by 40% through fine-tuned retrieval, chunking strategies, and evaluation feedback loops.
- Engineered an AI transcription service exposed via **REST/gRPC APIs** (85%+ accuracy); cut manual review effort by 50% and integrated outputs into downstream analysis pipelines — shipped end-to-end without hand-holding.
- Delivered across **Agile** sprints: requirements, design, code review, CI/CD via **Docker** and **TravisCI**; reduced post-release defects by 20% through systematic profiling and testing discipline.

Software Development Intern

Jul. 2023 – Oct. 2023

Humanify Technologies

Mumbai, India

- Built reusable **React Native** component utilities with **Jest** tests (90%+ coverage); cut duplicated code by 40% and accelerated form upload latency by 60% through algorithmic optimisation.

Software Development Intern

Feb. 2022 – May 2022

Analytics Vidhya

India

- Constructed a **Python** ML classification model end-to-end — data cleaning, EDA, feature engineering; delivered findings through data visualisations and executive stakeholder summaries.

PROJECTS

Beacon – City-Scale Emergency Response Platform | *Go, React 18, TypeScript, Flutter, GKE, PostgreSQL, Redis, ClickHouse,*

- Zero-to-one production platform: 12 repositories, 6 Go microservices on **GKE Kubernetes Autopilot** — event-driven via **GCP Pub/Sub** (CloudEvents 1.0, circuit breaker, exponential backoff); three-tier data layer (PostgreSQL CQRS, Redis cache, ClickHouse audit trail); **React 18 + TypeScript** ERT dashboard and **Flutter** mobile app. Live at beacon-tcd.tech.
- 13 **Apache Airflow** ETL pipelines ingesting live Irish public infrastructure data; **Vertex AI** Gemini Pro + Flash for real-time conversational triage and risk analysis; 13 external API integrations (Firebase, Twilio, TomTom, SendGrid) — distroless Docker images (~25MB), 4-role JWT-validated RBAC.

TrafficBook – Geo-Distributed Traffic Platform | *Python, FastAPI, React 18, TypeScript, PostgreSQL, Redis, RabbitMQ, Docker,*

- Built from scratch: 7 **Python FastAPI** microservices + **React 18** + **TypeScript** frontend deployed across **3 independent GCP regions** (EU/US/APAC); **BASE eventual consistency** via RabbitMQ federation — isolated PostgreSQL per region, no shared database.
- Ghost reservation pattern (two-phase **Redis** optimistic locking) for distributed race condition prevention; **Prometheus/Grafana** observability; **k6** load tested at 100 VUs — P95 <500ms, <5% error rate; **GitHub Actions** CI/CD.

BuilderX – SaaS Page Builder | *React, TypeScript, Node.js, PostgreSQL, REST API, Jest* **Explorify** – Adaptive Mobile App

- **BuilderX**: React + TypeScript drag-and-drop SaaS UI — virtual-DOM diffing (sub-50ms re-renders), Command-pattern undo/redo, Node.js REST API + PostgreSQL persistence, Jest tested, Lighthouse 94. **Explorify**: cross-platform React Native app (iOS + Android) adapting routes in real time based on user accessibility needs and environmental context.

EDUCATION

Trinity College Dublin

Dublin, Ireland

M.Sc. Computer Science – Future Networked Systems (Predicted First-Class Honours)

Aug. 2025 – Sept. 2026

- Modules: Distributed Systems, Scalable Computing, Machine Learning, AI, Security & Privacy, Advanced Software Engineering. Dissertation: *Energy-Efficient Cooperative Caching in Vehicular Edge Computing using Multi-Agent Deep Reinforcement Learning*.